



ENHANCING SALES VISIBILITY AND INVENTORY DECISION-MAKING IN A SMALL RETAIL STATIONERY BUSINESS

BDM CAPSTONE PROJECT

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ABOUT THE BUSINESS AND OBJECTIVES



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DATE: 12/02/26

Sl no	Time	Product category	Name of Product / company	Rate	Qty	Mode of Payment
1	10:05	clay	Kores clay activity kit	699	1	cash
2	11:00	Eraser	Apsara eraser	20	1	cash
3	11:50	Pouch	fancy pouch	330	1	UPI
4	12:15	Pen	Pilot VT pen	180	3	cash
5	12:58	copy gold	A4 copy gold	540	2	cash
6	01:30	Dairy	Yours Dairy	280	2	cash
7	01:55	Book	classmate spiral book	370	2	cash
8	02:30	Badminton	Badminton racket	4000	1	UPI
9	03:20	car	Metal car	750	1	cash
10	03:50	File	Expandable file	240	3	cash
11	04:35	Pen	Add Gel Roller pen	300	2	UPI
12	06:05	Blue book	Blue-book 32 pages	200	20	cash
13	08:30	Pen	Curve Ball pen	120	4	cash
14	09:10	Art kit	Done Art kit	398	2	cash
				4827		

Fig: Ledger for Sales data on 12.02.26

- Om Stationeries, Dattagalli, Mysuru
- Small neighbourhood retail shop
- 17 product categories, 664 SKUs
- Operations run entirely on owner experience – no structured records

PS1 – Profitability Visibility: The owner assumes a uniform 15–20% gross margin across all products, but has no structured way to verify this.

PS2 – Inventory Movement: Restocking relies on visual judgment, with no data to identify fast/slow-moving stock or capital lock-in.

PS3 – Experience vs. Data: All decisions rely on owner memory, with no records to verify or challenge them.

DATA COLLECTION

Sales Data:

- Store transactions recorded in a structured ledger and digitised into a spreadsheet
- Period: 10 Feb – 10 Mar 2026 (29 days, 458 transactions)

Inventory Data:

- Product list compiled with owner input; category mapping done using ChatGPT and verified
- Stock quantities based on owner estimates (14–16 Feb 2026)
- **No full physical count conducted to avoid operational disruption**

Sales Data

Metric	Value
Unique SKUs sold	173
Total revenue	₹1,47,216
Mean transaction value	₹321
Median transaction value	₹260
Std Dev (transaction value)	₹243
Min / Max transaction	₹15 / ₹1,475
Mean daily transactions	15.8
Payment split	Cash 58.1% / UPI 41.9%

Inventory Data

Metric	Value
Total SKUs	664 (657 priced)
Categories	17
Total estimated stock value	₹12,94,585
Mean selling rate	₹121
Median selling rate	₹50
Mean gross margin %	26.70%
Median gross margin %	30.00%
Std Dev gross margin %	6.40%
Min / Max margin %	0% / 40%
SKUs below 20% margin	159 (24.2%)
SKUs at/above 20% margin	498 (75.8%)



ABOUT DATA



Code	Category Name	SKUs
WI	Writing Instruments	57
PP	Paper Products	81
FF	Files and Folders	47
OD	Office Desk Supplies	43
AT	Adhesives & Tapes	33
CE	Correction & Erasing Items	21
MG	Measurement & Geometry Tools	29
AC	Art & Craft Supplies	88
SS	School-Specific Supplies	36
OF	Office Forms & Registers	26
PC	Printing & Computer Stationery	56
FB	Fasteners & Binding Materials	28
EM	Envelopes & Mailing Supplies	29
MU	Miscellaneous / Utility Items	22
TG	Toys & Games	14
EL	Electronics & Accessories	14
SV	Services	7
—	UNKNOWN (pending classification)	10

Sales Data

Column Name	Data Type
SI No	Integer
Date	Date
Time	String
Brand	String
Product Name	String
SKU	String
Rate (₹)	Numeric
Qty	Integer
Cost (₹)	Numeric
Mode of Payment	Categorical

Inventory Data

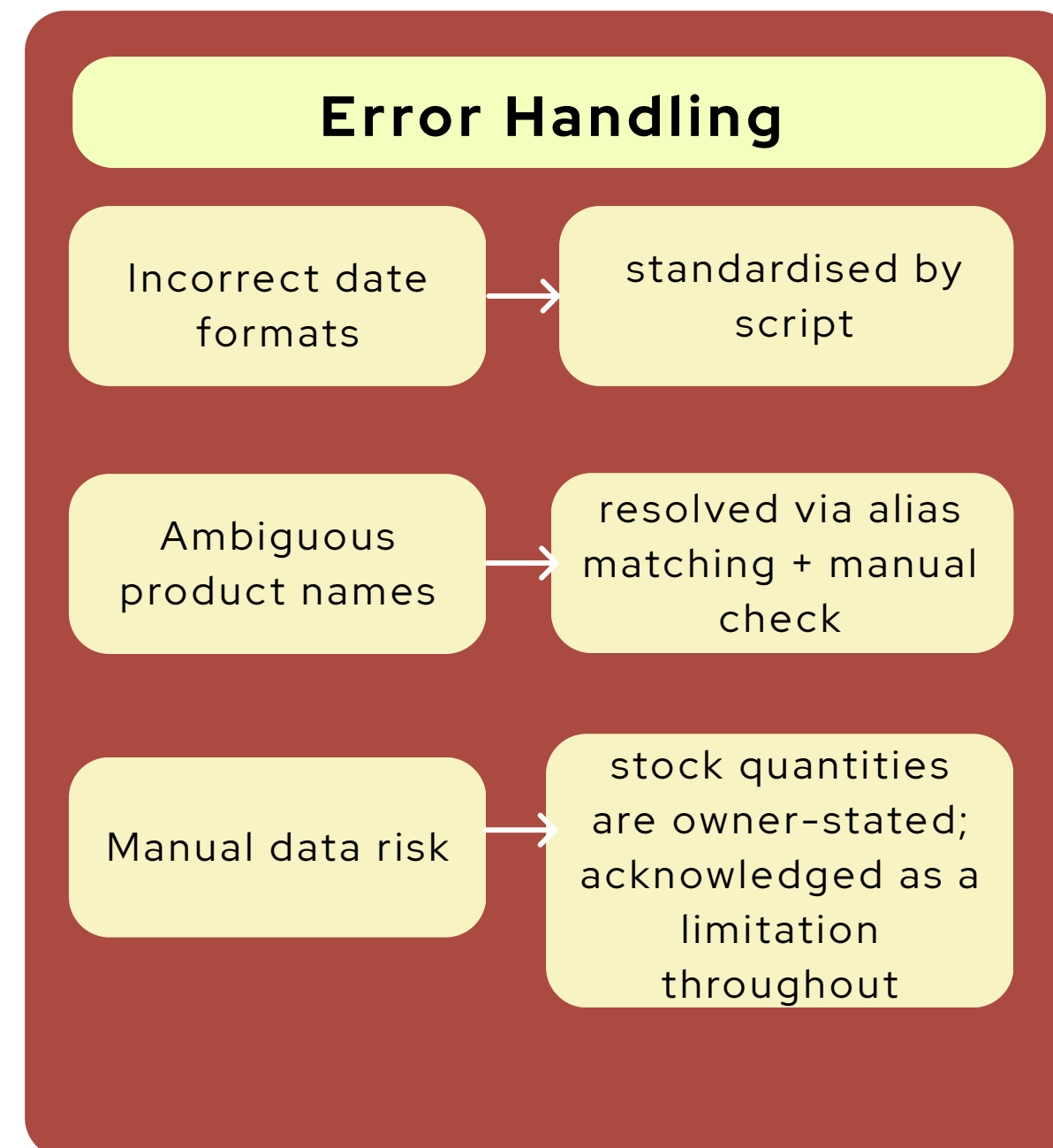
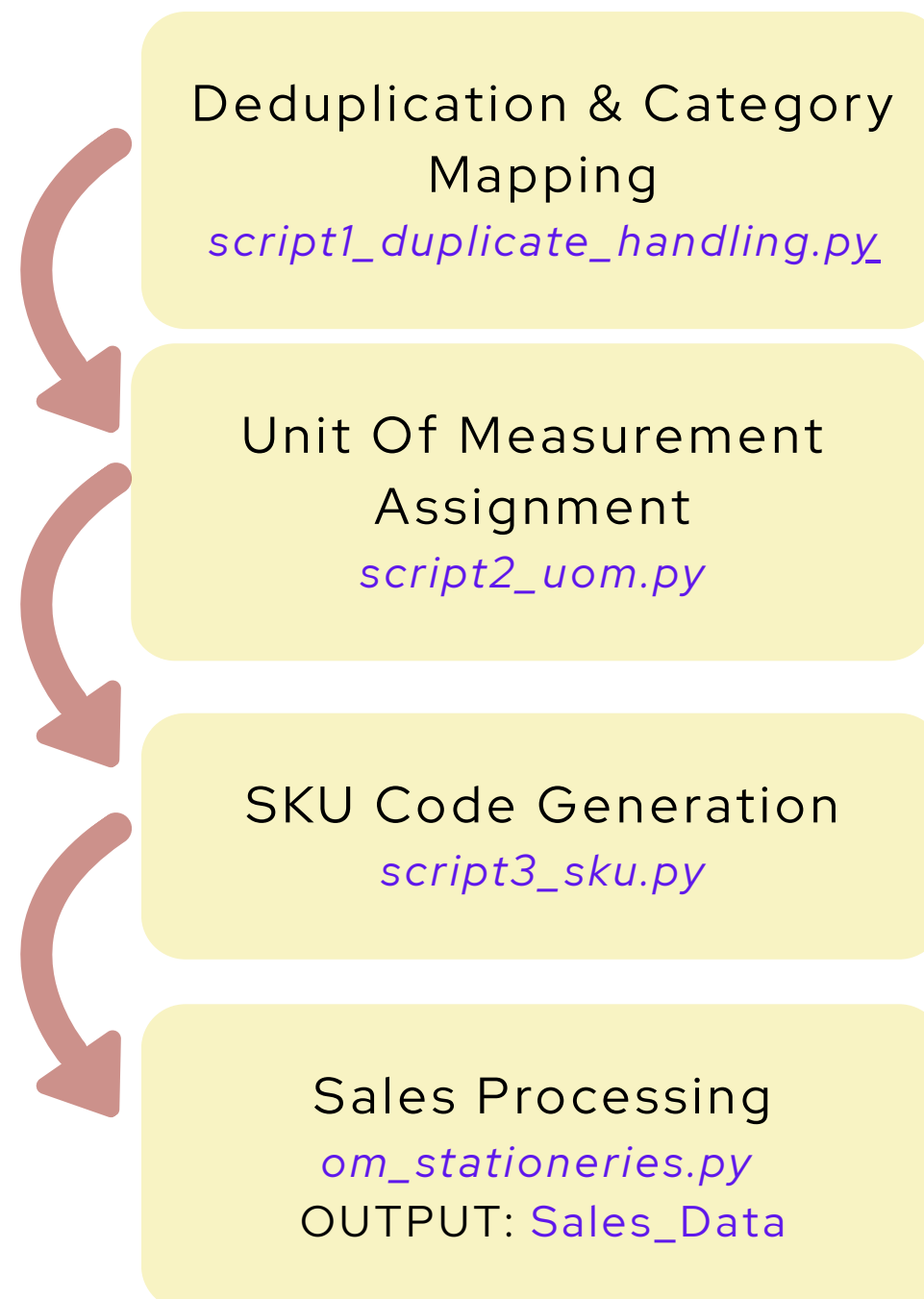
Column Name	Data Type
SL No	Integer
SKU Code	String
Category	Categorical
Sub-category	Categorical
Name	String
Unit of Measurement	Categorical
Rate	Numeric
Qty	Integer
Cost Price	Numeric
Aliases	String

DATA HANDLING

SKU Mapping:

[CATEGORY CODE] – [SUB-CATEGORY CODE] –[SEQUENCE NUMBER]

Example: WI-PEN-011

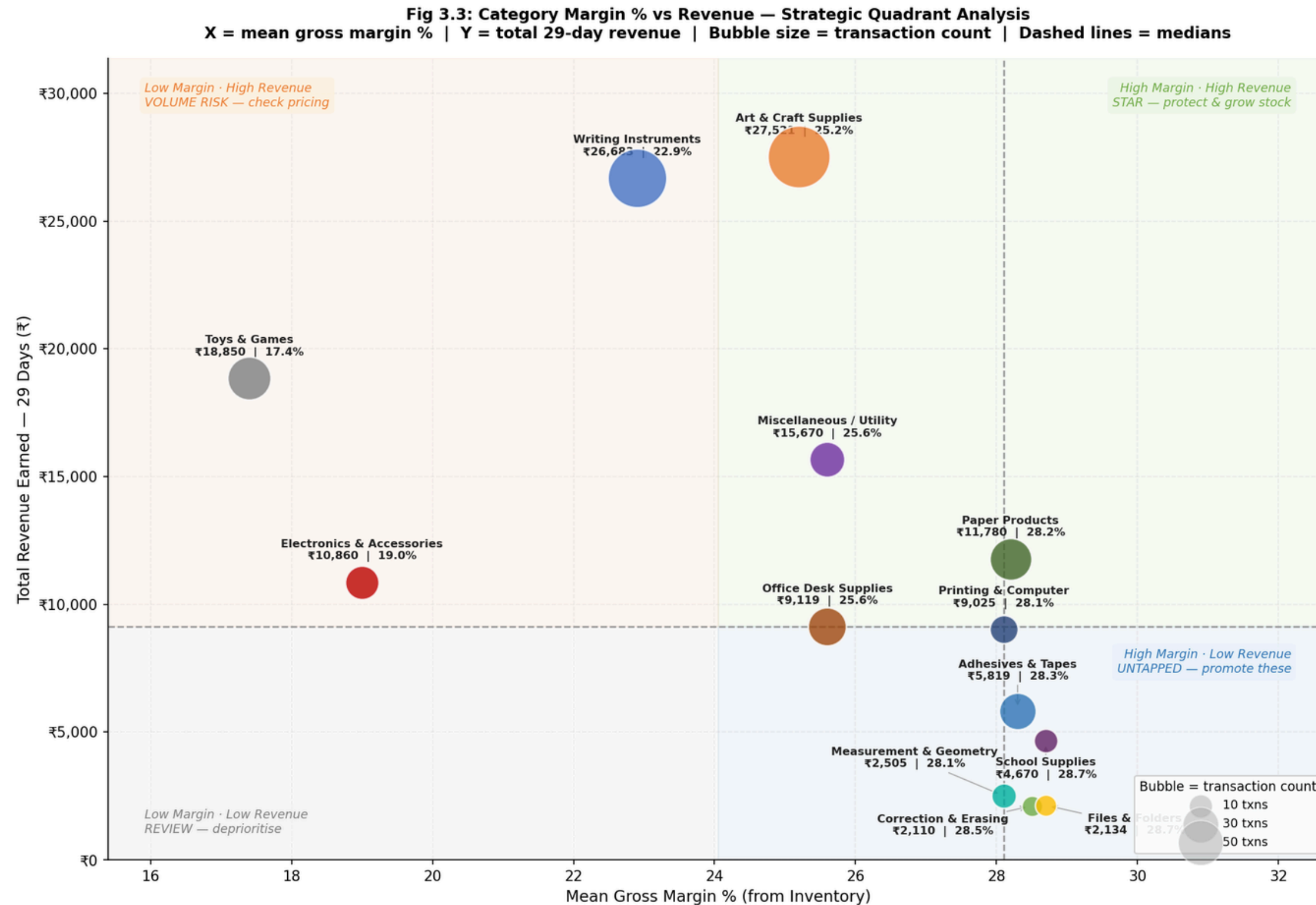


TOOLS

Python, Google Colab

- Mean vs Median margin (26.7% vs 30%) – The gap signals left skew. A subset of low-margin SKUs pulls the average down. The median is more representative of a typical SKU.
- Transaction value skew (₹321 mean vs ₹260 median) – Right skew from occasional high-value purchases. A typical customer visit is closer to ₹260.

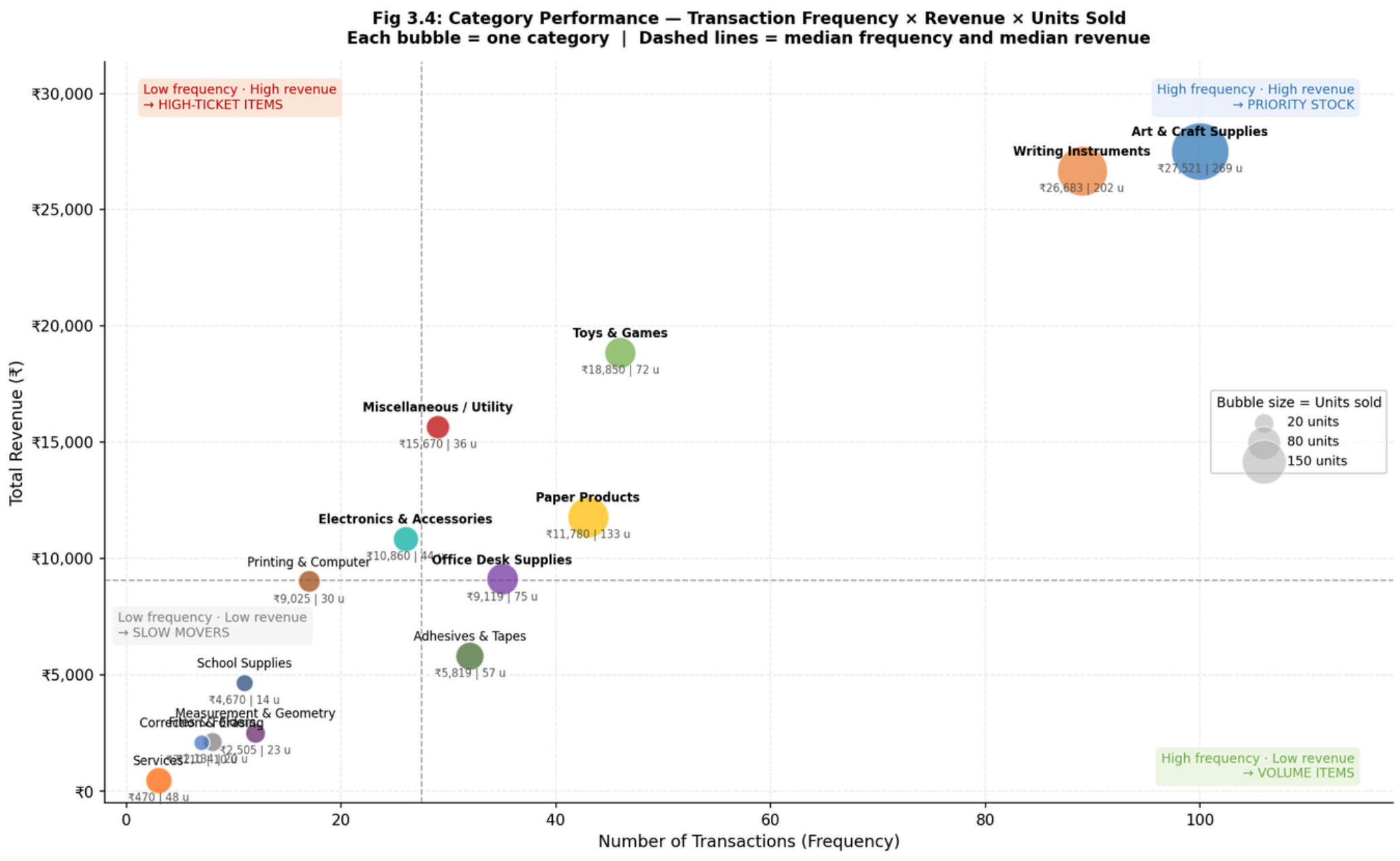
ANALYSIS



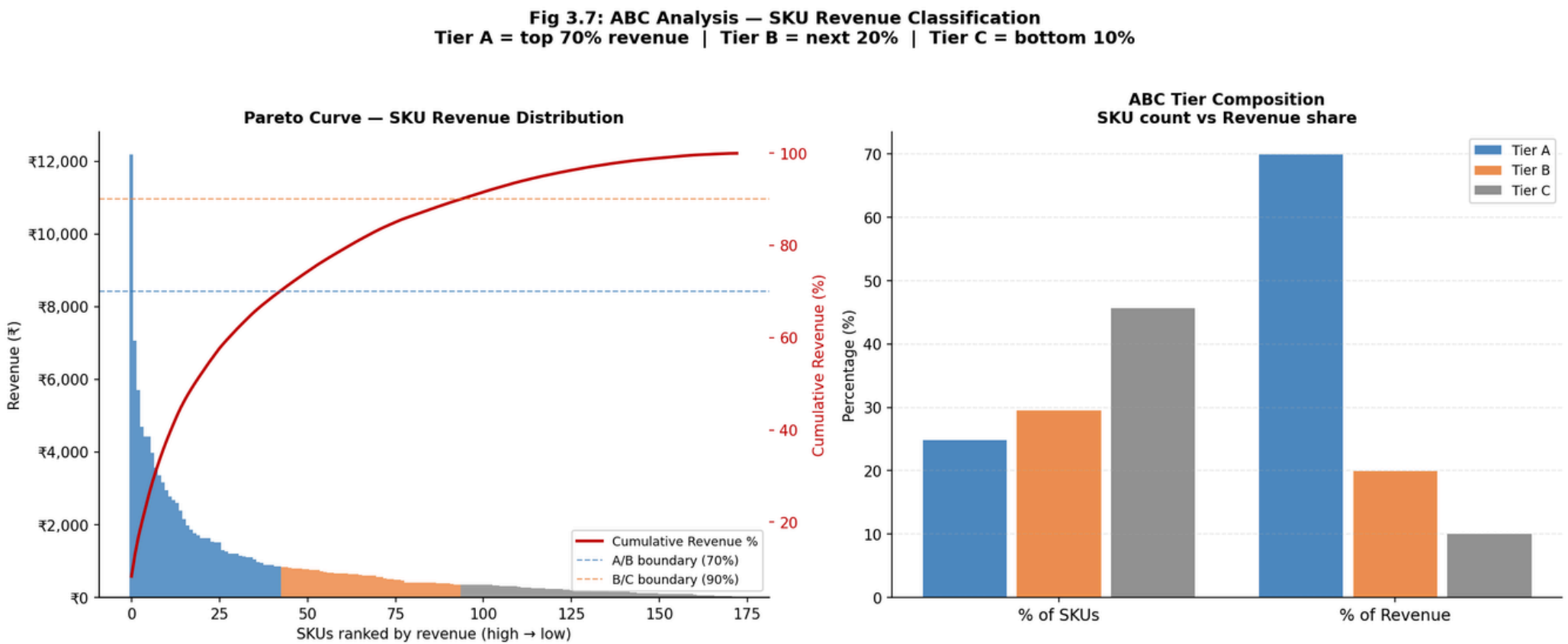
- 24.2% of priced SKUs fall below the 20% threshold — nearly 1 in 4 products is not meeting the owner's assumed margin floor
- Writing Instruments (₹26,683 revenue, 22.9% margin): 47% of its 58 SKUs are at or below 20% — this is a systemic category problem, not isolated outliers
- Art & Craft Supplies is the only top-revenue category operating above the median margin — it is the store's sole "Star"; all other high-revenue categories are in the Volume Risk quadrant



ANALYSIS

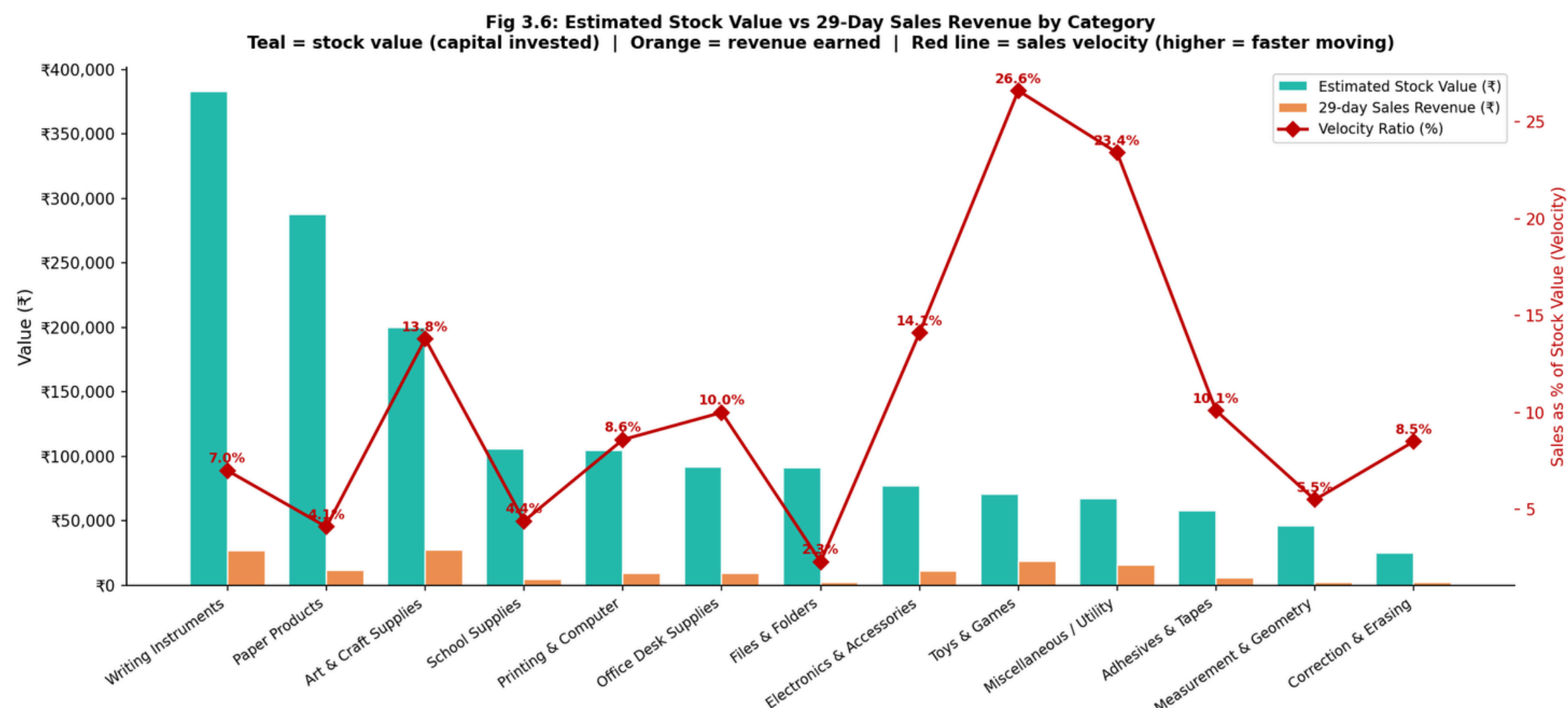
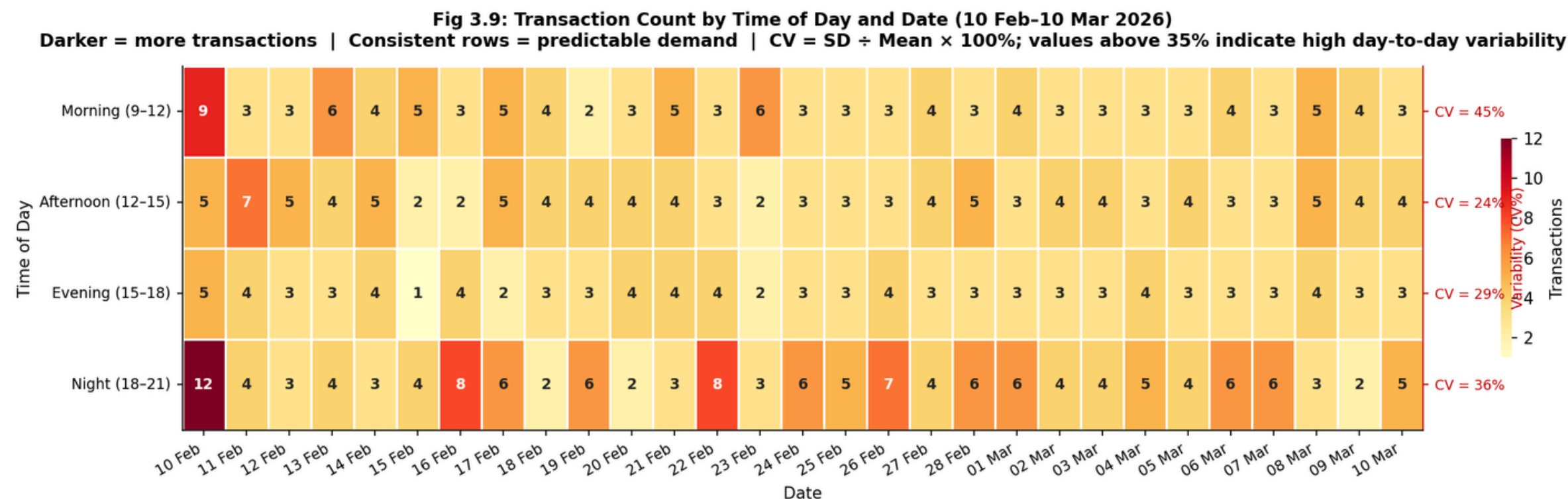


- Art & Craft Supplies and Writing Instruments lead on all three dimensions (frequency, revenue, units) – stockouts in either category directly impact daily revenue; these require weekly restocking checks
- 43 SKUs (25% of active products) generate 70% of total revenue – managing stock availability for these 43 products alone secures the majority of store income
- Writing Instruments holds ₹3,80,000 in estimated stock at 7% velocity – only ₹7 of every ₹100 invested converted to revenue in the study window; Files & Folders holds ₹89,000 at 2.3% velocity – the capital allocation is inverted relative to actual sales





ANALYSIS



- Night (CV = 40%) and Morning (CV = 36%) – the two highest-transaction slots – are also the most volatile; transaction counts swing from 2 to 12 in the same time slot, making intuition-based preparation unreliable
- Evening (CV = 24%) is the most predictable slot – operational decisions anchored to Evening patterns will be most reliable
- Acrylic paints (32 units, 17% margin), Staple pins (27 units, 17%), A4 copier paper (26 units, 18%) – the store's best-selling SKUs by volume are all priced below the margin floor; high sales volume amplifies the margin shortfall

INSIGHTS AND RECOMMENDATIONS



Findings

Recommendations

PS1: Pricing

24.2% of SKUs below 20% margin; top revenue categories (Writing Instruments 22.9%, Toys & Games 17.4%) are the most at-risk

- Immediate pricing audit of Writing Instruments and Toys & Games; prioritise the 9% of Writing Instrument SKUs in the 0–15% band
- Set category-level margin floors – not a single store-wide assumption
- A ₹2–5 price increase on top 5 high-volume low-margin SKUs (Acrylic paints, Staple pins, A4 paper, Wax crayons, Art kits) produces meaningful margin improvement at 25–30 units sold per month

PS2: Inventory

Writing Instruments: ₹3.8L stock, 7% velocity. Files & Folders: ₹89K stock, 2.3% velocity. Top 43 SKUs generate 70% of revenue.

- Tier A (43 SKUs): zero-tolerance stockout – check daily, restock at 2-week forward cover
- Tier B (52 SKUs): weekly monitoring, restock at 1-week cover
- Tier C (78 SKUs): monthly review; demand-signal ordering only
- Categories above 500 days of stock cover (School Supplies, Files & Folders, Correction & Erasing): do not reorder until stock drops below 180 days – frees an estimated ₹4–5 lakh in working capital

PS3: Operations

Night CV = 40%, Morning CV = 36%. UPI share swings 20–63%. High-volume SKUs priced below margin floor.

- Introduce a 5-minute daily log at close of day: transaction count per time slot + Cash vs UPI revenue – this single habit generates all data needed for PS2 and PS3 decisions going forward
- When Night average exceeds 6 transactions for 3 consecutive days: ensure full stock and change float

THANK YOU